

July 23rd, 2026

SCHC Architecture Discussion

Side meeting

Draft : draft-ietf-schc-architecture-06

Authors : A. Pelov , P. Thubert , A. Minaburo , Q. Lampin , M. Dumay

IETF 126, Vienna



Note Well

By participating in the IETF you agree to follow IETF processes and policies. This Note Well is a reminder of some of those policies. For a linked version of this text, please visit www.ietf.org/note-well or use the QR code below.

- IETF participants are expected to behave in a professional manner and extend respect and courtesy to their colleagues at all times (see *RFC 7154: IETF Guidelines for Conduct and IETF Anti-Harassment Policy*). If you have any concerns about behavior, please contact the *Ombudsteam* who have a duty of confidentiality and extensive powers to act, as set out in *RFC 7776: IETF Anti-Harassment Procedures*.
- If you are aware that any IETF contribution (as defined in *RFC 5378: Rights Contributors Provide to the IETF Trust*) is covered by patents or patent applications that are owned or controlled by you, your employer or your sponsor, you must disclose that fact, or not participate in the discussion (see *RFC 8179: Intellectual Property Rights in IETF Technology*).
- For detailed process information consult *RFC 2026: Internet Standards Process* and *RFC 2418: IETF Working Group Guidelines and Procedures* and updates to those.
- The IETF routinely makes public written, audio, video, and photographic records of IETF activities, including your personal information as set out in the *IETF Privacy Statement*.

For advice, please talk to Working Group chairs or Area Directors.



Agenda

- Introduction and outline – 30 min
 - A little bit of context
 - Objectives and scope of the Architecture document
 - Impacts of -06 on existing documents
 - Current definitions of core components
 - Endpoints, Instances, Stratum, ...
 - A few scenarios to illustrate those definitions
- Discussion – 1h



History and Objectives

- Objectives
 - Document SCHC
 - For generic use (including LPWANs)
 - Provide unified terminology
 - Clarify and fill gaps
 - Informational document
 - Cover simple, as well as more advanced use-cases
 - Document the work of the WG
 - Work should happen on standard-track documents
- Evolution 2021 -> 2026
 - Work 2021-2025 (lpwan-> schc)
 - Side meeting during IETF123
 - Design team creation after IETF124
 - Open to anyone to join, consensus-driven
 - Work presented during interim meeting on May 12th, 2026



How do the drafts work with the architecture?

SCHC Framework Core + Extensions

[RFC 8724 SCHC Framework](#)

[RFC 9441 Compound Ack](#)

[draft-ietf-schc-universal-option-01](#)

[draft-corneo-schc-compress-payload-02](#)

[draft-corneo-schc-ctx-mgmt-00](#)

[draft-westerlund-schc-compute-address-00](#)

SCHC Interoperability

[RFC 9363 YANG Data Model](#)

[draft-ietf-schc-schclet-00](#)

[draft-ietf-schc-protocol-numbers-06](#)

[draft-lampin-voici-02](#)

[draft-pelov-schc-header-format-00](#)

SCHC Adapted to

[RFC 9011 LoRaWAN](#)

[RFC 9391 NB-IOT](#)

[RFC 9442 Sigfox](#)

[draft-ietf-schc-over-networks-prone-to-disruptions-03](#)

[draft-ietf-schc-over-ppp-00](#)

SCHC Applied to

[RFC 8824 CoAP](#)

[draft-ietf-ipsecme-diet-esp-10](#)

[draft-ietf-schc-icmpv6-compression-02](#)

[draft-ietf-schc-8824-update-09](#)

[draft-sirohi-schc-quic-frame-compression-00](#)

[draft-ietf-schc-architecture-06](#)

Informational

- Document SCHC (Generic: LPWAN and other uses)
- Clear terminology
- Cover simple, as well as more advanced use-cases
- Entry-point for non-SCHC people (KISS)



Take-away from today

How -06 impacts existing documents

- Automatically processed 7 RFCs and 17 drafts against -06 version (Claude, Gemini, meta-analysis with Codex)
- ✓ For all 7 RFCs - all concepts, roles, communication patterns, rule structures, and fragmentation mechanisms map directly or compositely to SCHC architecture -06
- ✓ All 17 drafts map onto the architecture with High or Very High conceptual equivalence
- ✓ 14 Easy, 3 Medium, 0 Hard - the bulk is terminology alignment, not technical rework
- With diff suggestions
 - <https://github.com/lp-wan/archi/tree/main/archi-06-document-analysis>
- The work continues



The Contents

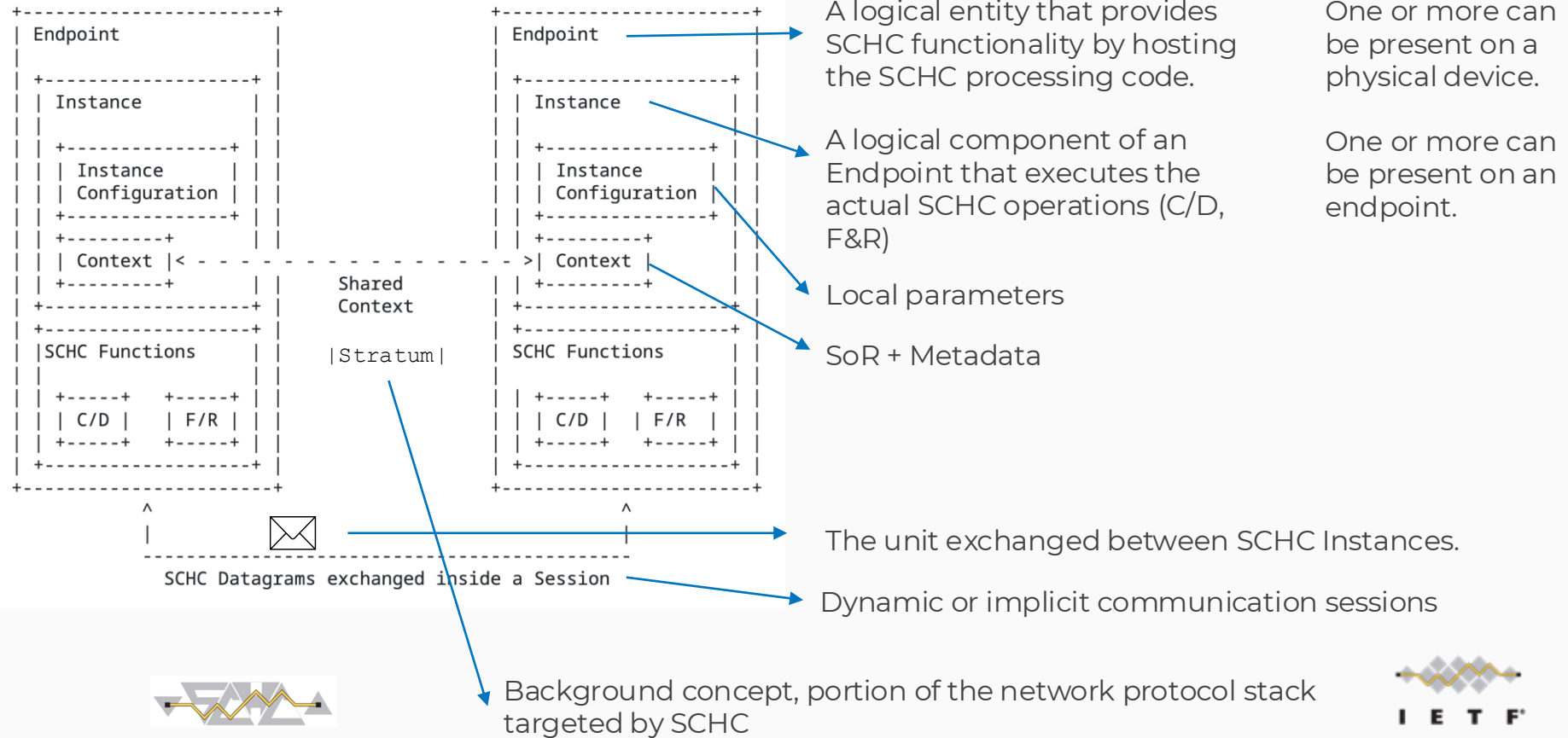
A new structure designed to introduce the main entities and their relationships right from the first few pages

1.	Introduction	3
2.	Requirements Language	4
3.	Terminology	4
4.	Architecture	6
4.1.	Overview of a Basic Architecture	6
4.1.1.	SCHC: Quick Reminders	6
4.1.2.	Basic SCHC Architecture	6
4.2.	Focus on core components	9
4.2.1.	Instance	10
4.2.2.	Endpoint	11
4.2.3.	Session	13
4.2.4.	Domain	14
4.2.5.	Datagram Format	15
5.	Deployment Profiles	16
6.	Operational considerations	16
6.1.	Error handling	17
6.2.	Context consistency	17
7.	Security Considerations	17
8.	IANA Considerations	18
9.	References	18
9.1.	Normative References	18
9.2.	Informative References	18

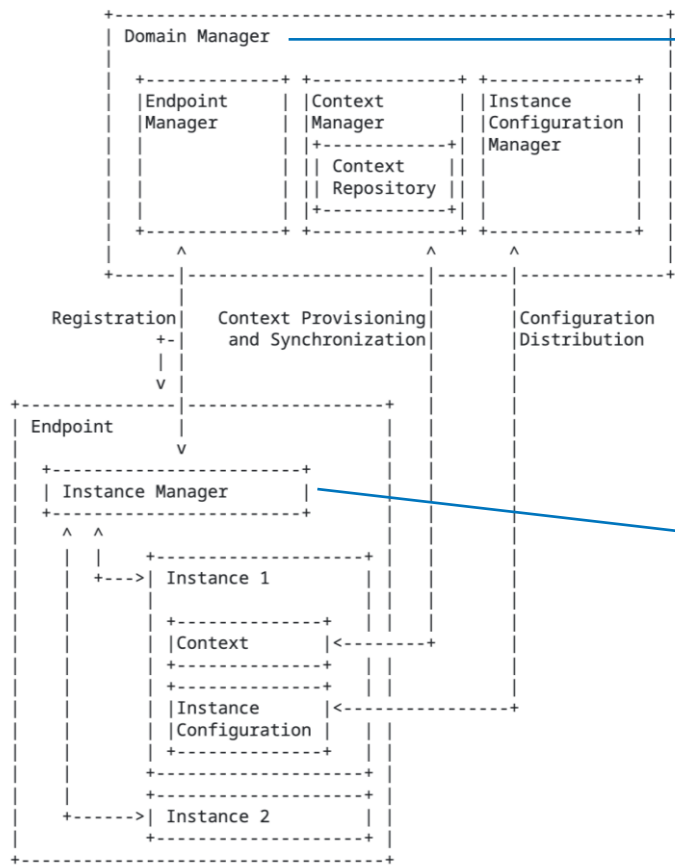
Appendix A.	Examples - To Be Validated	20
A.1.	Control Header Examples	20
A.2.	Deployment Models	21
A.2.1.	LPWAN deployment	21
A.2.2.	PPP deployment	25
A.2.3.	Direct transport over Ethernet, IPv6, and UDP	26
A.3.	Compatible Partial Contexts	27
Appendix B.	Future Work - To Be Decided by the Working Group	28
B.1.	C/D Engine Interface	28
B.2.	Other Open Items	28



Basic SCHC Architecture



Domain



Domain: A logical grouping of Instances that share a common Context.
Domain Manager: manages the **Domain**, including Context synchronization and configuration distribution.

It uses:

Endpoint Manager to manage the **Endpoints** (TBD)

Context Manager to manage the **Contexts**

Instance Conf Manager to manage **Instance Configuration**

Domain vs. Session

The Domain defines which Instances share Contexts, while the Session identifies which Instances are actually communicating.

A logical component that manages the lifecycle and configuration of Instances within an Endpoint.

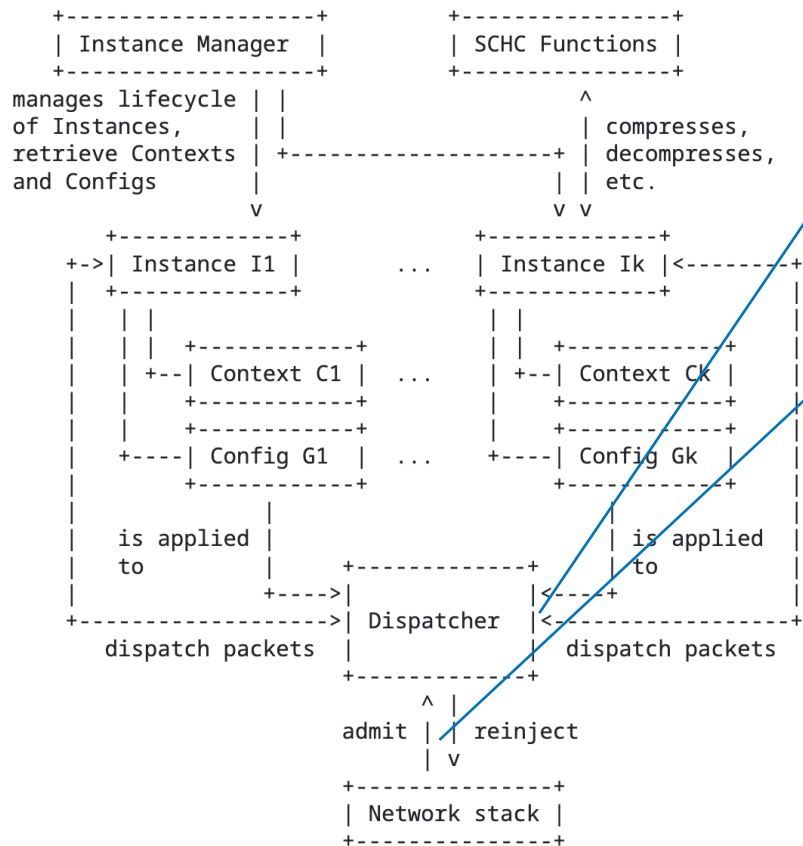
E.g. creating, updating, and deleting Instances as needed.

Note: These are logical components.

They may be implemented in a single software for example.

The domain may be peer-to-peer.

Multiple Instances on an Endpoint



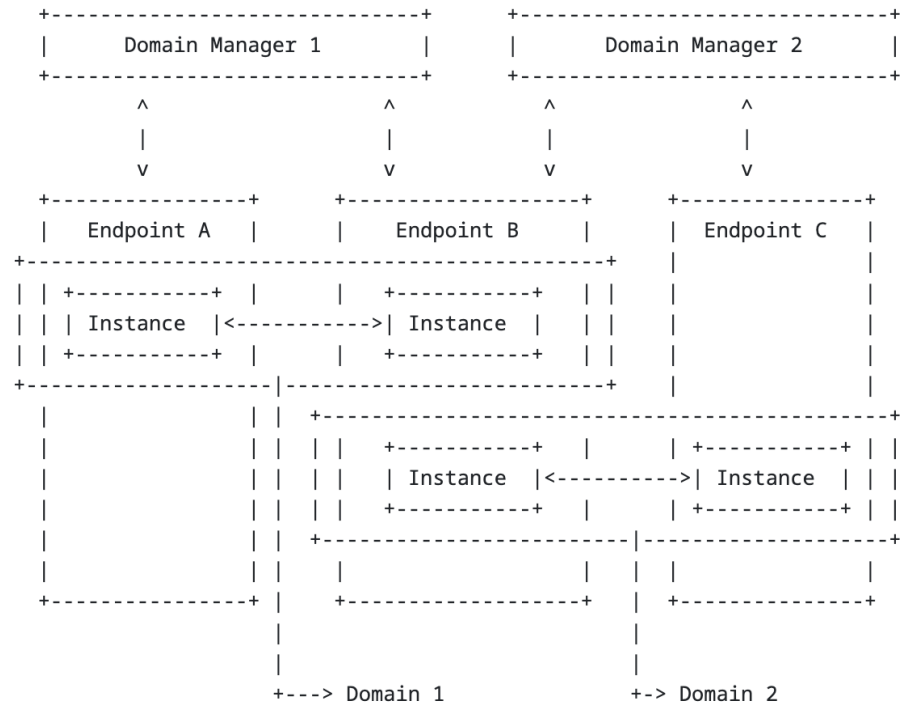
Dispatcher: A logical component of the Endpoint that routes Datagrams to the appropriate Instances based on defined admission rules.

It can be integrated into the network stack or implemented as a separate component.

Discriminator: An optional information element used by the Dispatcher to route SCHC Datagrams to the appropriate Instance. The discriminator can be a combination of several criteria.

Multi-Tenants LPWAN Scenario

1 Gateway, 2 IoT devices with different Contexts



1 Gateway, 2 IoT devices with different Contexts

A single SCHC Instance runs on each IoT Endpoint (A, C).
Two Instances run on the Gateway (B)

- Single Domain per IoT Endpoint
- Two Domains on Gateway
- Discriminator is provided by L2, e.g., DevEUI.



SCHC Datagram Format

Datagram Format

```
+-----+-----+
| RuleID | Rule-dependent fields and payload |
+-----+-----+
```

A Datagram starts with a RuleID. The Rule identified by that RuleID determines the format and interpretation of the remaining bits. A Datagram may be an unfragmented SCHC Packet or a SCHC F/R message, such as a SCHC Fragment, SCHC ACK, SCHC ACK Request, SCHC Sender-Abort, or SCHC Receiver-Abort.

Control Header for Advanced Use Cases

```
+-----+-----+-----+
| RuleID | Control Header | Rule-dependent fields and payload |
+-----+-----+-----+
```

or

```
+-----+-----+-----+
| Control Header | RuleID | Rule-dependent fields and payload |
+-----+-----+-----+
```

The presence, placement, and format of the Control Header must be clearly identified, e.g., by a SCHC profile document or other specification that defines the framing used by the deployment.

Example of a control header for generic multiplexing use-cases is given in: [draft-lampin-voici-02](#)

It is expected there to be several such standard-track documents that specify different control headers.



SCHC Header and SCHC Header Format

- SCHC Header: The portion of a SCHC Datagram that carries SCHC processing information, including the RuleID and, when present, Control Header or F/R control fields. The SCHC Header enables the receiver to identify the Rule and interpret the remaining SCHC Payload, fragment, residue, or control information. Its exact encoding is defined by the applicable Rule, profile, or encapsulation.
- SCHC Header Format: A profile-, encapsulation-, or deployment-specific description of the SCHC Header encoding, including the presence, order, and encoding of the RuleID and any Control Header or F/R control fields.

Example of how this can be implemented: [draft-pelov-schc-header-format-00](#)



Thank you!

- Discussion
- Questions
- Next steps

